

Low-Frequency Ambient-Noise Measurements in the South Fiji Basin

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Abstract—The effect of wind speed on ambient noise has been measured in an experiment carried out in the South Fiji Basin. The noise data in the band 15–250 Hz were well correlated with the variations in the local wind speed. The relationship between noise level N and wind speed v is expressed by $N = B + 20n \log v$. The constants B and n have been estimated by fitting the data using this model. Our analysis indicates that there are two types of behavior: For $v > 15$ kn, a value of $n = 1.5$ was obtained for the entire band, whereas for $v < 15$ kn there is no correlation with wind speed observed in the data. Also, the results suggest that there is a delay of 40–120 min for the effect of wind on the hydrophone noise level.

I. INTRODUCTION

IT IS well established from research in underwater acoustics that the level of ambient ocean noise is related to the local wind speed. There has recently been considerable interest among physical oceanographers and underwater acousticians in the processes of noise generation at the sea surface, and in the physical parameters that control the noise level. Experiments carried out by Medwin and Beaky [1], and Farmer and Vagle [2] point to the importance of bubbles in breaking waves as the source of wind-generated noise at the surface. These experiments have, however, been concerned primarily with frequencies greater than 1 kHz.

We present in this paper new results for the relationship between the noise level and wind speed at low frequencies, from 15–250 Hz. The noise measurements were obtained in an experiment using a towed line array to study the level and the directionality of ambient noise in the South Fiji Basin. We report direct measurements of the omni-hydrophone and array-beam noise level in the open ocean where the wind speed was measured *in situ*. The noise levels at frequencies in the band from 15–250 Hz are well correlated with wind speed, and our data suggest that there are two regions of behavior, depending upon the wind speed.

The following sections will describe the experiment and the data processing, and will discuss the measured data. The results for the relationship between noise level and wind speed are compared to other results reported in the literature.

II. EXPERIMENT

The data were obtained in an experiment carried out during July 1982 at a site in the South Fiji Basin (Fig. 1).

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The basin is an acoustically isolated region of the southwest Pacific which provides an excellent natural laboratory for studying noise processes in the ocean. It is bounded by deep-sea ridges and islands in the west and east, by the New Zealand land mass in the south, and by the Fiji Plateau in the north. The region of depth excess for surface sound sources occurs only inside the 4000-m contour shown in Fig. 1. However, the amount of depth excess is small and convergence zone propagation is weak. Outside the basin, acoustic propagation is bottom-limited.

In addition to the bathymetric isolation, the ambient-noise background in the South Fiji Basin is low compared to sites in the northern oceans, because the shipping density is very small. The merchant shipping was monitored using ship weather-report data that was reported during the time of the exercise. From this information, the tracks and positions of all ships passing through the region were known. For a typical 24-h period, there were fewer than two ships in the region.

The noise measurements were made with a horizontal line array which was towed by the research ship HMNZS *Tui* for a period of 12 days in the vicinity of the site. The array depth (nominally 400 m) was monitored at the ends and the mid-point, and the system was towed at a speed of 2 kn. The data were digitized at a rate of 1 kHz and the spectra were estimated typically for 1 min averages in 1-Hz bins over the band 0–255 Hz. A cross-spectral beamformer which utilized a discrete Fourier transform was used to carry out beamforming on the cross-spectral averaged fields [3]. In addition to the acoustic data, wind speed was recorded at 5-min intervals on the ship during the exercise, and the sound speed profile in the water was measured with expendable bathythermographs and a CTD probe.

III. AMBIENT NOISE DATA

Typical omnidirectional power spectra measured during the sea trial are shown in Fig. 2 for wind speeds of 5–8 kn (lower curves) and for 30 kn (upper curve). The spectra in this figure were obtained about 15 h before and then during the passage of a storm over the site (16–18 July), when the wind speed increased from 5 to 35 kn. The strong lines at 60 and 90 Hz are from the tow ship, and the abrupt increase in level below 10 Hz is due to array self-noise. It is significant to note that the background noise was essentially flat over the band from 15 to 250 Hz. This is clearly demonstrated in Fig. 2 for the high wind-speed data. The hump in the spectrum from 120–180 Hz, which is evident in the data for lower wind speeds, is due to biological activity [4]. It is also

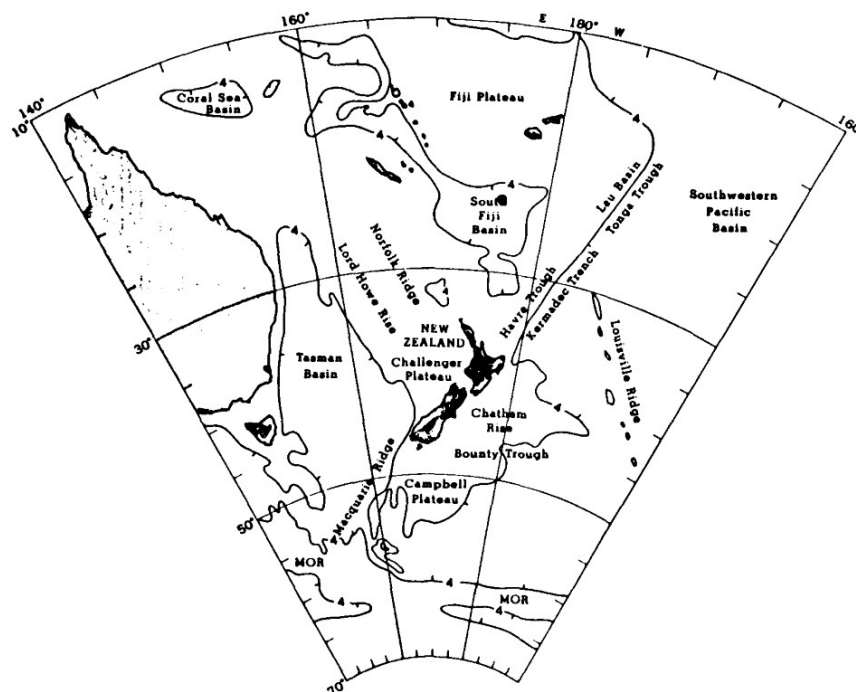


Fig. 1. Location of the experiment in the South Fiji Basin. The 4000 m contour roughly defines the boundary of the basin.

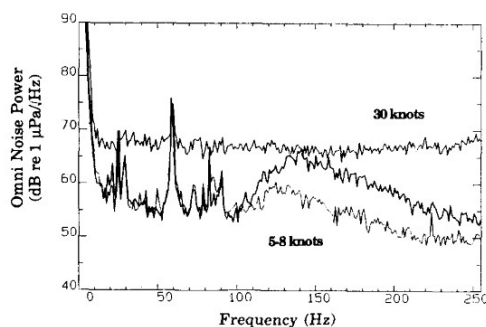


Fig. 2. Omni-hydrophone noise spectrum levels (dB re $1 \mu\text{Pa}/\sqrt{\text{Hz}}$) versus frequency for wind speeds of 5–8 kn (lower curves), and 30 kn (upper curve). These data were recorded before and during the passage of a storm over the site, 16–18 July.

obvious that the noise levels measured in this experiment are very low compared to the spectrum levels from sites in the northern oceans; at frequencies below 50 Hz, the difference is as much as 25 dB for similar wind conditions [4].

An overall view of the variation in ambient noise levels over the duration of the experiment is provided in Fig. 3 for frequencies of 18, 145, and 250 Hz. These data represent 1-min averages of the noise taken at 10-min intervals, and averaged in 5-Hz bands centered at each frequency; the gaps in the data are due to equipment malfunctions. Also plotted is the local wind speed measured on-board the tow ship.

The noise levels for the 250-Hz band can be seen to follow the wind speed quite closely. The occasional spikes in the

data are associated with system transients in the wet-end electronics and usually occur when the array is undergoing a course change. The 18-Hz band also shows a reasonable correlation with wind speed, particularly during the period 16–18 July, when a storm passed over the site. In addition, there are a large number of spikes that are associated with noise from seismic events in the region. A detailed analysis of the acoustic T-phase components of the seismic transients from some of these events has been reported previously [5]. The temporal variation for the 145-Hz band presents a contrast to the other two bands shown in the figure. For this band, there is very little correlation between the single hydrophone (omni) noise spectra and local wind speed. Instead, daytime levels are reasonably constant at around 62–64 dB, whereas nighttime levels can drop abruptly by as much as 10 dB. There is some evidence of a correlation between the nighttime level and local wind speed using the beam-formed data, and a discussion of this result will be presented in the next section. The cause of the diurnal variation in this band has been attributed to biological activity [4].

IV. WIND-GENERATED NOISE

The relationship between the omni-noise level and wind speed is shown Fig. 4, where the values for the 250-Hz noise level from Fig. 3 have been plotted against local wind speed. While there is an obvious correlation that is most clearly demonstrated at high wind speeds (greater than 15 kn), there is also considerable scatter in the data. At wind speeds less than 5 kn there is no obvious trend, and the data are likely

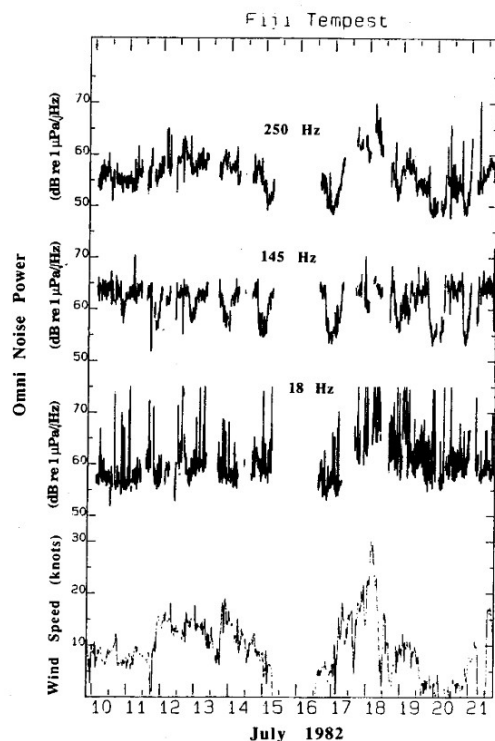


Fig. 3. Omni-noise levels (dB re 1 $\mu\text{Pa}/\sqrt{\text{Hz}}$) at 18, 145, and 250 Hz measured at 10 min intervals during the experiment. The wind speed measured on-board the ship is shown at the bottom of the figure.

limited by electronic system noise. Following the method of Piggott [6] and Crouch and Burt [7], we have fitted the data to a model given by:

$$N = B + 20n \log v \quad (1)$$

where N is the noise level in dB, v is the wind speed, and n and B are constants. If a linear regression is utilized over all data points, the estimated values for n and B are 0.44 and 48 dB, respectively. This result is a poor fit for the higher values of v , where one would expect a strong correlation with wind speed. Evidence that the "true" value of n is likely to be larger is provided by limiting the regression to utilize data points with wind speeds greater than 15 kn. In this case, there is a good fit for the high values of wind speed, and the resulting values for n and B are 1.5 and 23.6 dB, respectively. This regression is shown by the line in Fig. 4.

We have followed the latter approach in estimating values for n and have separated the data into two regimes of wind speed. A value of 15 kn was chosen as the separation point. This approach is based on the consideration that any noise sources other than wind that are present in the data will increase the noise level at a given wind speed. We assume that this effect will significantly increase the measurements for low wind speeds, and consequently will lower the value of n . The noise estimates for high wind speeds are less likely to be affected by extraneous (nonwind) noise sources because the relative contribution of wind noise is greatest for high

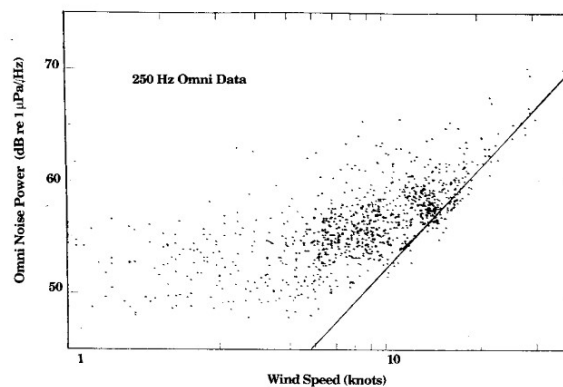


Fig. 4. Noise levels (dB re 1 $\mu\text{Pa}/\sqrt{\text{Hz}}$) versus wind speed for the 250-Hz band. The line shows the regression for wind speeds > 15 kn.

TABLE I
ESTIMATES OF n , B , AND r DERIVED FROM OMNI-NOISE DATA

Omni Hz	$v > 15$ knots			$v < 15$ knots		
	n	B	r	n	B	r
18	1.91	14.3	0.51	0.00	60.0	0.00
30	1.42	28.4	0.60	0.00	62.3	0.01
45	1.80	16.1	0.72	0.00	58.5	0.06
75	1.46	25.1	0.76	-0.01	59.4	-0.09
100	1.39	25.5	0.72	0.00	57.6	-0.06
125	0.91	39.9	0.54	0.00	61.4	-0.06
145	0.86	40.7	0.43	-0.01	61.6	-0.12
180	1.05	34.4	0.60	0.01	59.1	-0.08
250	1.46	23.6	0.80	0.00	55.4	0.05

TABLE II
ESTIMATES OF n , B , AND r DERIVED FROM BEAMFORMED DATA

Beam Hz	> 15 knots			< 15 knots		
	n	B	r	n	B	r
18	1.76	12.8	0.58	0.01	55.0	0.18
30	2.13	2.3	0.61	0.02	52.9	0.25
45	1.57	17.3	0.52	0.03	53.4	0.28
75	1.83	9.1	0.64	0.01	49.7	0.14
100	1.98	6.1	0.75	0.01	50.0	0.11
125	1.49	20.3	0.74	0.01	53.5	0.07
145	1.48	20.5	0.76	0.00	54.6	0.00
180	1.45	20.4	0.70	0.00	52.6	0.00
250	1.32	25.4	0.71	0.01	53.1	0.07

values of v . It follows that the lowest measured noise levels for any given wind speed are the most significant values in our analysis. An indication that our assumptions are reasonable is given in Fig. 2 by comparing the spectra for the low and high wind speeds. At the high wind speed, for instance, the noise due to biological activity is not evident in the spectrum.

The results of the regression for various frequencies from 18–250 Hz are summarized in Table I. The parameter r is the correlation coefficient calculated for the data points used in the regression. For high wind speeds, the lowest values of n are those obtained at frequencies of 125, 145, and 180 Hz. The data for these frequencies are contaminated with biological noise, which has produced higher noise levels than would be expected for low-to-medium wind speeds, thus lowering the slope of the regression line. The low value of r for these frequencies indicates that the estimates for n and B are less reliable.

Table II shows the results of the same process applied to beamformed noise data. The lowest beam levels for a particu-

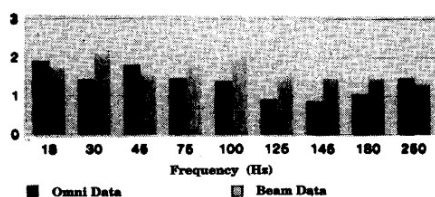


Fig. 5. Bar graph summary of the estimates of n over the band 15–250 Hz.

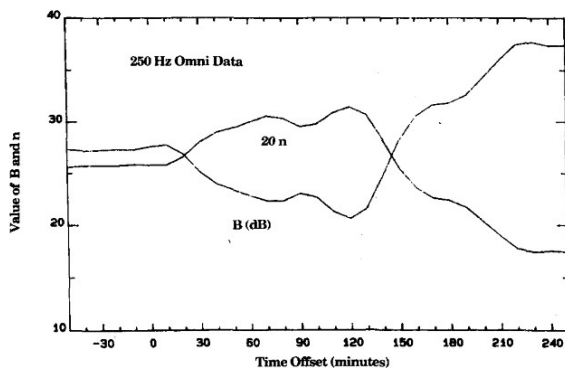


Fig. 6. Values of B and n determined for various delays between the times of the noise level and wind-speed measurements. The results for delays from 40–120 min have the highest correlations.

lar noise estimate were used in the regression. Considerably increased correlation is achieved for the 125–180-Hz band, and the estimated values for n and B are closer to those obtained for the 250-Hz data. The estimates of n for all the frequencies are summarized in the bar graph in Fig. 5. A value $n = 1.5$ is consistent over the entire band for high wind speeds, whereas there is no correlation with wind speed for speeds less than 15 kn.

Fig. 6 shows the variation of the estimated values of n and B at 250 Hz for different offsets between the times of the wind-speed measurements and the associated omni-noise measurement. Better correlation and higher values of n are obtained if a delay of 40 min to 2 h is used. This result suggests that there is a similar delay in the effect of local wind on hydrophone noise levels. The tabulated results shown above were obtained using a delay of about 40 min.

V. DISCUSSION

Our analysis of the relationship between noise level and wind speed at low frequencies suggests that there are two regimes of behavior, depending on the wind speed. At low wind speeds, below 15 kn, there is no correlation between noise level and wind speed observed in the data, whereas at high wind speeds there was good correlation over the entire band from 15–250 Hz. The value of 15 kn defining the boundary between the two types of behavior was chosen to be slightly greater than the critical wind speed for the generation of whitecaps, about 10 kn [8]. Thus it was ensured that

breaking waves existed at the sea surface for the high wind speed regime in our analysis. This result supports the suggestion reported by Carey and Browning [9] that there may be two mechanisms for the generation of sea-surface noise. At high wind speeds, the most likely mechanisms are related to the bubbles created by breaking waves [8], [9]. At low wind speeds, in the absence of whitecaps, other mechanisms such as capillary wave/wave or nonlinear wave interactions [10], [11] may be important.

There have been several other estimates of the value of n reported in the literature, and the results have been summarized by Carey and Browning [9]. Most of the experiments have been carried out in the northern oceans and the data are often contaminated by shipping noise. The bulk of the data for high wind speeds that are not dominated by shipping noise yield values for n in the range $0.9 < n < 2.2$. Other data reported by Burgess and Kewley for the Indian Ocean [12] and the data of Shooter and Gentry [13] for a near-bottom hydrophone in the equatorial Pacific indicate values of around 1.5. The latter data were, however, dominated by shipping noise sources below about 125 Hz. Our results suggest a value of $n = 1.5$ for the entire low-frequency band from 15–250 Hz.

VI. SUMMARY

We have presented results for the relationship between noise level and wind speed, based on new measurements of ambient noise at low frequencies. The data were obtained using shallow hydrophones in the open ocean in an experiment where the wind speed was measured *in situ*. Our analysis suggests that there are two types of behavior, depending on the wind speed. At high wind speeds, greater than 15 kn, the noise estimates were well correlated with wind speed, with a value of $n = 1.5$ for the band from 15–250 Hz. At lower wind speeds, there was no correlation with wind speed. In addition, there is evidence of a time delay of at least 40 min for the effect of wind on the hydrophone noise levels.

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